AI IMPACT ON STUDENTS

KEY INSIGHTS REPORT

50,000 Student Records | Data Analysis by Pritesh Pujari

50,000	17	8	5
Students	Features	Key Insights	Majors

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DATASET OVERVIEW

50,000	+0.203	87.5%	8.4 hrs	25.0%	75.8%
Total Students	Avg GPA Change	GPA Improved	Avg AI hrs/wk	High Burnout	Avg Skill Retention

KEY INSIGHTS

■ OPPORTUNITY

Insight 1 — AI Helps Academic Performance, But Not Equally

- 87.5% of students improved their GPA after using AI tools
- Average GPA improvement: +0.203 points across all 5 majors
- STEM students benefit the most (+0.217), followed by Medical (+0.201)
- Business students see the smallest gain (+0.194) — likely due to use case mismatch

+0.203 avg GPA change across 50,000 students

Major	Avg GPA Change	Burnout (High %)	Interpretation
STEM	**+0.217**	30.0%	Best gains, highest risk
Medical	**+0.201**	23.2%	Strong gains, moderate burnout

Major	Avg GPA Change	Burnout (High %)	Interpretation
Humanities	**+0.198**	20.7%	Consistent, lowest burnout
Arts	**+0.197**	22.7%	Stable performance
Business	**+0.194**	24.3%	Lowest gain, mid burnout

■ RISK

Insight 2 — STEM Students Have the Highest Burnout (30%)

- 30% of STEM students are in the High Burnout category
- Humanities shows the lowest burnout rate at just 20.7%
- Burnout is not driven by skill level — Advanced (25.5%), Beginner (24.6%), Intermediate (24.9%)
- High AI hours combined with high academic pressure is the likely cause in STEM

30% of STEM students — High Burnout Risk

■■ WATCH

Insight 3 — Institutional Policy Has a Bigger Impact Than Expected

- Strict Ban schools: lowest GPA gain (+0.187) AND highest burnout (29.8%)
- "Allowed With Citation" policy yields the best academic outcomes (+0.208)
- "Actively Encouraged" performs similarly (+0.206) with only 23.8% burnout
- Banning AI increases student stress without any academic benefit

Strict Ban = +0.187 GPA change vs +0.208 with citation policy

■ OPPORTUNITY

Insight 4 — HOW You Use AI Matters More Than How Much

- Debugging / Troubleshooting: highest GPA gain at +0.249
- Ideation and Copywriting: moderate gains around +0.200
- Direct Answer Generation: lowest gain at only +0.133
- Students who problem-solve with AI outperform those who use it as a shortcut

Debugging (+0.249) vs Direct Answers (+0.133)

Primary Use Case	Avg GPA Change	Category
Debugging / Troubleshooting	**+0.249**	Best
Copywriting / Drafting	**+0.200**	Average
Ideation	**+0.200**	Average

Primary Use Case	Avg GPA Change	Category
Summarizing Reading	**+0.197**	Average
Direct Answer Generation	**+0.133**	Lowest

■■ RISK

Insight 5 — Heavy AI Use Hurts Skill Retention

- No AI usage: Skill Retention = 77.6% (highest)
- Light usage (0-5 hrs/wk): 76.0% retention
- Moderate usage (5-15 hrs/wk): 77.0% — sweet spot for balance
- Heavy usage (15-40 hrs/wk): retention drops to 72.7% (lowest)

Heavy AI users score 4.9% lower on skill retention

AI Usage Band	Hours / Week	Skill Retention %	Avg GPA Change
No Usage	**0 hrs**	**77.6%**	**+0.262**
Light	**0–5 hrs**	**76.0%**	**+0.195**
Moderate	**5–15 hrs**	**77.0%**	**+0.227**
Heavy	**15–40 hrs**	**72.7%**	**+0.173**

■ WATCH

Insight 6 — Traditional Study Still Beats AI Hours for GPA

- Correlation: Traditional Study Hours vs GPA Change = +0.376 (strong)
- Correlation: GenAI Hours vs GPA Change = -0.047 (near zero)
- AI is a supplement — it does not replace the benefit of focused human study
- Students who combine both (traditional + moderate AI) perform best

r = +0.376 (traditional) vs r = -0.047 (GenAI)

■ WATCH

Insight 7 — Paid Subscriptions Provide Only a Marginal Edge

- Paid users: avg GPA change = +0.207 | Skill Retention = 75.4%
- Free users: avg GPA change = +0.200 | Skill Retention = 76.1%
- Difference in GPA is minimal (+0.007); free users actually retain more
- Conclusion: the tool matters less than how students use it

+0.207 (Paid) vs +0.200 (Free) — only 0.007 gap

■ RISK

Insight 8 — 4.7% of Students Are Critically At Risk

• 2,330 students have both High Burnout AND High AI Dependency (score >= 7)

• AI Dependency is strongly correlated with burnout risk (r = +0.360)

• These students are likely over-reliant on AI and struggling academically

• Institutions must identify and provide targeted interventions for this group

2,330 high-risk students — need urgent institutional intervention

RECOMMENDATIONS

For Institutions	For Students	For Educators
• Shift from Strict Bans to "Allowed With Citation" policy — data shows better outcomes • Identify the 4.7% high-risk students and offer counselling + workload support • Invest in AI literacy workshops to move students from passive to active AI use	• Use AI for debugging and ideation — NOT to generate direct answers • Keep 5–15 hrs/week as the sweet spot; heavy use reduces skill retention • Never replace traditional study with AI — correlation data is clear (r = +0.376)	• Design assessments that reward problem-solving, not recall — AI-resistant tasks • Monitor STEM students closely — highest performers but highest burnout too • Track AI usage patterns; Prompt Skill Level predicts GPA outcomes significantly

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